

Your recent learnings

When you left 3 days ago, you worked on Aggregating DataFrames, chapter 2 of the course Data Manipulation with pandas. Here is what you covered in your last lesson:

You learned about creating and manipulating pivot tables in pandas, a powerful feature for summarizing and analyzing data. Pivot tables allow you to calculate grouped summary statistics, similar to what you can achieve with spreadsheet software. Here are the key points you covered:

- **Creating Pivot Tables:** You discovered how to use the `.pivot_table()` method in pandas to group data and calculate summary statistics. For instance, to calculate the mean weight of dogs by color, you would specify the `values` as the column to summarize and the `index` as the column to group by.
- **Custom Summary Functions:** You learned that the default aggregation function is the mean, but you can use the `aggfunc` argument to apply other functions, like NumPy's `median` function, for different summary statistics.
- **Multiple Summary Statistics:** By passing a list of functions to `aggfunc`, you can obtain multiple statistics at once, such as mean and median weights.
- **Grouping by Multiple Variables:** You explored how to group data by more than one variable by passing a second variable name to the `columns` argument, allowing for more detailed analysis.
- **Handling Missing Values:** The `fill_value` argument was introduced as a way to replace missing values (NaNs) with a specified value, such as zero, to avoid gaps in your data analysis.
- **Adding Margins:** You saw how setting `margins=True` adds additional rows and columns to the pivot table, showing the mean of all values in a row or column, providing a high-level summary of the data.

Here's an example code snippet from the lesson:

```
import pandas as pd
import numpy as np

# Example of creating a pivot table
pivot_table = df.pivot_table(values='Weight', index='Color', aggfunc=np.median)
```



This lesson equipped you with the skills to aggregate and summarize data efficiently, making it easier to derive insights from your datasets.

The goal of the next lesson is to teach you how to efficiently navigate and manipulate pandas DataFrames using indexing and slicing techniques.

Was this useful? ☒ yes ☐ no

Got it